DS605: Fundamentals of Machine Learning
Academic Year 2026-27 (Autumn)

Instructor:	Dr. Arpit Rana (arpit_rana@dau.ac.in)
Prerequisites:	Programming in Python (Ref.: Python for Programmers by Deitel & Deitel 2021 – Ch. 1 to 10)
Slot:	M.Sc. Data Science I Semester
Category:	Core
Course Credits:	4 Credits (3-0-2)
Lectures (LAB A SF):	Tuesday (15:00 - 16:00) | Thursday (14:00 - 15:00) | Friday (16:00 - 17:00)
Lab Session (LAB A SF):	TBD
TA contact info:	TBD

Course Description:

This course provides a comprehensive introduction to the fundamentals of machine learning, which are essential for first-year M.Sc. Data Science students. The curriculum covers classical machine learning techniques, including supervised and unsupervised learning, decision trees, support vector machines, clustering, and outlier analysis. Students will learn the foundational concepts of learning frameworks, model evaluation, and validation. The course emphasizes practical applications and hands-on experience through projects and lab sessions, preparing students to understand and implement basic machine learning algorithms and methodologies.

Suggested Books:

- Data Mining and Machine Learning: Fundamental Concepts and Algorithms by Zaki and Meira, 2nd Edition, Cambridge University Press [2020].
- Machine Learning with Python Cookbook by Kyle Gallatin and Chris Albon, 2nd Edition.
- Understanding Machine Learning: From Theory to Algorithms by Schwartz and David [2022]

Course Outcomes:

After completing the course, the student should be able to -

- Understand the foundations of Machine Learning (ML).
- Develop skills in using recent ML techniques to solve practical problems.
- Gain experience in doing an independent study and research
- Apply all of the above in Python (using Python ML Stack: Scikit-learn, Numpy, Pandas)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X				X			X

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Evaluation Scheme:

Ability to Understand and Apply Theoretical Concepts		Ability to implement and Solve Real Problems and Present the Solution		
In-Sem Exams (25%) [Individual Contribution]	**End-Sem Exam (25%)** [Individual Contribution]	**Assignments (20%)** [Individual Contribution]	**Challenges (15%)** [Individual Contribution]	**Project (15%)** [Group Contribution]

Tentative Course Plan (this course plan is subject to change without notice.)

Units	Topics	# Lectures
Introduction	Definition, objectives, components, approaches, challenges, and case studies. Data: Structured, Semi-structured, and Unstructured. Algebraic, Geometric, and Probabilistic View of Data	3
Data Analysis Foundations	Data objects and attribute types. Measures of central tendency, dispersion, and proximity.	4
Data Preprocessing	Data Cleaning, Data Integration, Data Reduction, Data Transformation, and Discretization	2
Fundamentals of Predictive Analytics (supervised learning framework)	(i) Representation: hypothesis class—bias-variance trade-off and expressiveness-model complexity trade-off; (ii) Evaluation: consistency and generalizability; data splits; and measures of model performance; and (iii) Optimization: learning as a search: hyperparameter and parameter tuning;	9
Predictive Tasks	Regression and classification (binary, multiclass, and multi-label); Lazy learners: kNN-based regression and classification. Eager learners: Linear regression. Regularization and Polynomial regression. Logistic regression; Class-imbalance: Undersampling, Oversampling, and SMOTE	9
Classification Techniques	Linear and Non-linear Support Vector Machines: Classification using Hinge Loss; SVR. Decision Trees; Dimensionality Reduction using PCA and SVD; Ensemble Techniques: Voting Classifiers, Bagging and Pasting, Random Forests, Boosting;	9
Cluster Analysis	Basic concepts, Partitioning-based methods, Hierarchical methods, and Density-based methods; Evaluation of Clustering	6

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Preliminary Schedule of the Course:

Week	Lecture	Lab	Assignments/ Project
Week-1 [20-07-2026]	Introduction and Data Analysis Foundations	– No lab –	-
Week-2 [27-07-2026]	Data objects and attribute types. Measures of central tendency, dispersion, and proximity	Getting familiar with Git and Python Refresher	Group formation and domain finalization through a Google form
Week-3 [03-08-2026]	Fundamentals of Predictive Analytics-I: Representation	Data Scraping using **`Scrapy`** and Data Preprocessing	**Lab Assignment-1:** Data Scraping and Preprocessing
Week-4 [10-08-2027]	Fundamentals of Predictive Analytics–II: Evaluation	Data Wrangling with **`Pandas`**; Array-Oriented (Vectorized) Programming with **`NumPy`**	**Lab Assignment-2:** Vectorized Programming
Week-5 [17-08-2026]	Fundamentals of Predictive Analytics–III: Optimization	Getting familiar with the **`Scikit-learn`**: Data loading to Model's Performance Evaluation	**Lab Assignment-3:** Data Preprocessing using Scikit-learn
Week-6 [24-07-2026]	Lazy Learners: Regression Techniques	100-Minute Development Challenge	
Week-7 [31-08-2026]	First In-Semester Exams: 31 Aug to 5 Sept 2025 (Monday to Saturday) – No First In-Sem Exam of DS605: Fundamentals of Machine Learning –		
Week-8 [07-09-2026]	Dimensionality Reduction	Creating a Pipeline in Scikit-learn for an End-to-End ML Project	**Lab Assignment-4:** End-to-End Model Development using **`Scikit-learn`**
		100-Minute Development Challenge	
Week-9 [14-09-2026]	Linear Support Vector Machines	Handling Text (Sentiment Analysis) using Classical ML: **`NLTK`**	
Week-10 [21-09-2026]	Non-linear Support Vector Machines and Regressor	Handling Images (MNIST and F-MNIST) using Classical ML	

Week-11 [28-09-2026]	Decision Tree Classifier and Regressor	100-Minute Development Challenge	
Week-12 [05-10-2026]	Second In-Semester Exams: 6 Oct to 10 Oct 2026 (Tuesday to Saturday) – DS605: Fundamentals of Machine Learning Exam will be Conducted –		
Week-13 [12-10-2026]	Ensemble Techniques	Ensembling using **`Scikit-learn`**	**Lab Assignment-5:** End-to-End Ensemble Model using **`Scikit-learn`**
Week-14 [19-10-2026]	Ensemble Techniques Contd...	100-Minute Development Challenge **Course Project Announcement** **(Submission Deadline: 15-11-2026)**	
Week-15 [26-10-2026]	Cluster Analysis–I: Partitioning-based methods, Hierarchical methods;	100-Minute Development Challenge	
Week-16 [02-11-2026]	Cluster Analysis–II: Density-based methods; Evaluation of Clustering	Cluster Analysis using **`Scikit-learn`**	**Lab Assignment-6:** Clustering using **`Scikit-learn`**
Week-17 [09-11-2026]	In-Semester Break: 09 Nov to 13 Nov 2026 (Monday to Friday)		
Week-18 [16-11-2026]	**Course Project Evaluation**		
Week-19 [23-11-2026]	End-semester Examination: 23 Nov to 02 Dec 2026 (Monday to Next Wednesday)		

Guidelines for Lab Assignments, Development Challenges, and Course Project

Lab Assignments

- Each student must select a domain of personal interest and collect a large dataset related to it.
- Additional data scraping is required to enrich the dataset for model-building during the semester.
- By the **second week**, students must register their **domain of interest** and **team members**.
    - **Note:** No two team members can select the same domain.
    - Students from different teams may work on similar domains, but **no two students may use the same dataset**.
- Each lab session will be structured as follows:
    - The **first hour** will include a TA-led concept explanation and demo.
    - Assignments will be released after the demo.
    - Students will have **2 hours** to complete and submit their work (Colab/Jupyter notebooks) on Google Classroom.
- **Submissions must be made from within the M.Sc. Data Science lab only**. Remote submissions are not permitted.
- **Late submissions** will be penalized:
    - Submissions beyond **6 hours** of assignment release will receive **no credit**.
- A total of **five lab assignments** will be conducted during the course.

Development Challenges and Course Project

- Students will form teams of **two members**. Each team will:
    - Participate in **five timed ML Development Challenges** (each lasting **100 minutes**).
    - Complete an **End-to-End Course Project** based on one of the domains previously worked on individually: **define their project problem**, gather relevant data, and develop a solution.
- **Team evaluation will account for 15%** of the total course marks. Choose team members carefully.
- Teams will remain **fixed for the entire semester**; no changes will be allowed after registration.
    - With 60 students, there will be **30 teams** (one team may have five members).
    - Team registration will open in **Week 2** and remain open for **three days**. Registration will be via a Google Form.
- During the challenges, **relative grading** will be used based on the highest-performing team.

Course Project Details

- Each team will choose **one thematic domain**, such as:
    - E-commerce, Finance, Healthcare, Pharmaceuticals, Sports, Entertainment, Renewable Energy, Oil & Gas, Automobile, Agriculture, FMCG, Security, Social Media, Supply Chain, or any other relevant and engaging domain.
- Each group must:
    - Define a **unique problem** within the selected domain.
    - Use **reliable, preferably public, data sources**.
    - Ensure **no other group** is working on the **same dataset within the same domain**.
    - Enrich the dataset, if possible, by collecting additional data.
    - Every team member must be able to explain all aspects of their course project (concepts, code, methodology, etc.).

Submission and Evaluation

- One group member must submit the **final Course Project** on **Google Classroom or GitHub Classroom**.
- Evaluation will be conducted through a **Chalk Talk**, which includes:
    - Problem definition
    - Code review
    - Dataset scale and quality
    - Tools and technologies used
    - Methodology
    - Novelty and originality of the approach
    - All group members should be ready to answer any questions related to the project.
- **Late project submissions** (within 24 hours of the deadline) will incur a **20% penalty**.

Plagiarism Policy

Plagiarism will result in **zero marks** for Lab Assignments, Challenges, and the Course Project. Plagiarism includes:

- Copying code from any external source and submitting work not written personally by the student.
- Any suspected violation will be strictly penalized.